NASA C MAPSS FD001 Audit and Improvement

Leakage-controlled grouped validation, model comparison, and competition demo readiness

Prepared 20 October 2026 | CPU-bounded local execution

Executive conclusion

A five-fold engine-grouped pseudo-censoring audit was completed without loading official test files during selection. The validation champion is HGB with a 100-cycle training cap and causal 10/30-cycle features. It achieved RMSE 7.576, MAE 5.390, R2 0.946, and PHM08 mean 0.985 across 500 endpoint cases from 100 held-out engines.

This result applies to the predeclared 10-to-100-cycle validation horizon. It does not establish a general production cap above 100 cycles. The cap-125 HGB remains the wider-life reference, and official-benchmark results are reported separately without feeding back into selection.

1 Data and protocol audit

Data	Rows	Engines	Result
FD001 train	20,631	100	26-column schema and all integrity checks passed
FD001 test	13,096	100	Endpoint alignment verified
Validation	500 cases	100	5 GroupKFold folds x horizons 10/25/50/75/100

Features use only current and past cycles within each engine. Engine ID, final life, hidden suffixes, and official-test residuals are never input features. Cycle and cycles_seen were tested as a controlled, available-at-inference feature family; removing both degraded the cap-125 control.

2 Validation comparison

Candidate	RMSE	MAE	R2	PHM08 mean
hgb_cap100_w10_30	7.576	5.390	0.946	0.985
hgb_cap125_near_constant_removed	9.964	6.820	0.907	1.891
hgb_cap125_initial_delta	10.037	6.922	0.905	1.897
equal_blend_hgb_extra_trees	9.904	6.768	0.908	1.941
hgb_tuned_small_leaves	10.101	6.884	0.904	2.031
hgb_control_cap125_w10_30	10.063	6.799	0.905	2.032

The cap-100 model won validation. The cap-125 near-constant-sensor and initial-delta variants were close alternatives. ExtraTrees and the predeclared OOF equal blend did not surpass the validation champion. CNN-LSTM remains the strongest compact neural comparison from the earlier study, but its validation RMSE was higher than the tree models.

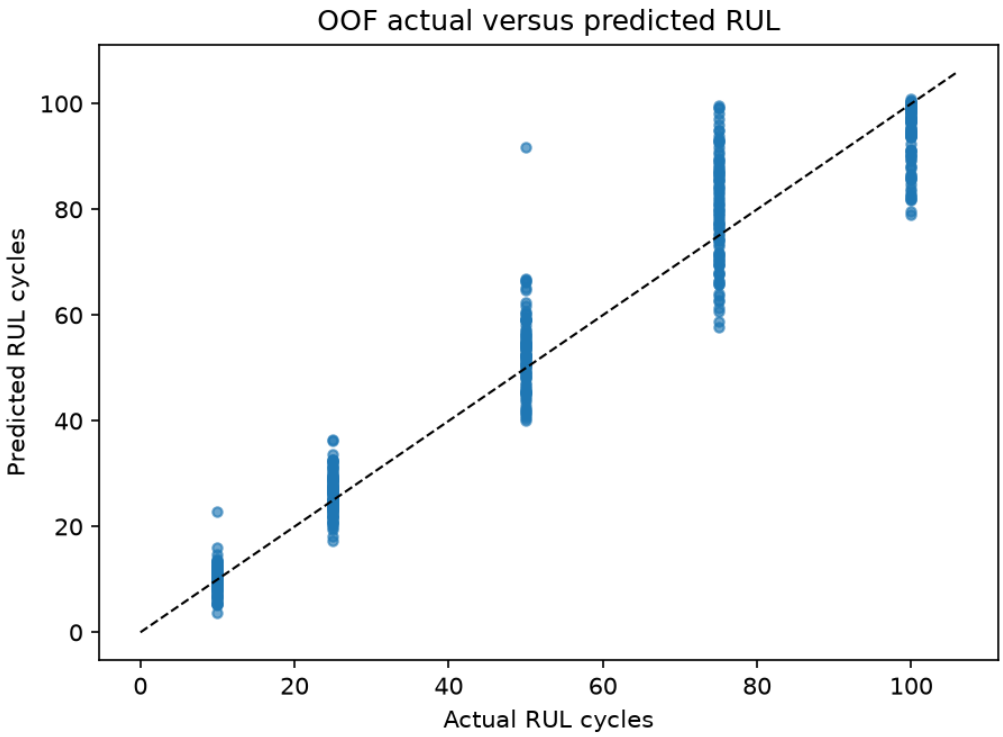


Figure 1. Out-of-fold actual versus predicted RUL for the validation-selected candidate.

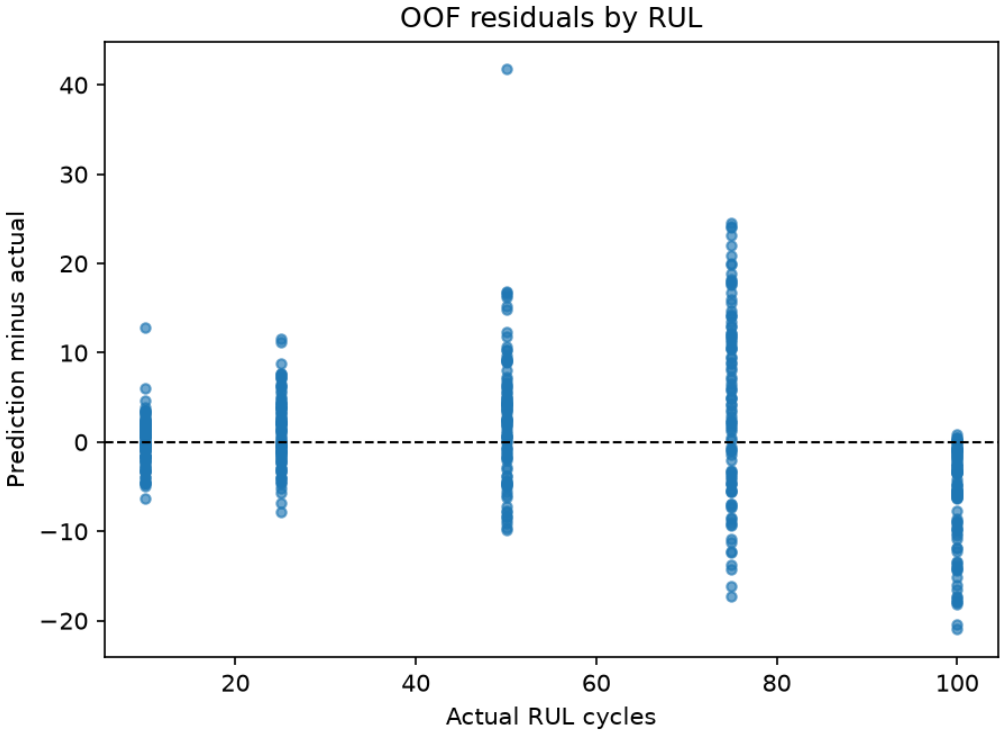


Figure 2. Out-of-fold residuals by true RUL. Values below zero are underpredictions.

3 Locked historical official benchmark comparison

After validation selection was locked, the cap-100 candidate was evaluated once on the already-inspected official FD001 benchmark. It is a secondary historical comparison and did not influence the feature, target, or model choice.

Metric	Historical benchmark value	Interpretation
RMSE	16.755 cycles	Lower is better
MAE	11.959 cycles	Lower is better
R2	0.837	Higher is better
PHM08 mean	3.393	Lower is better
Bias	-7.330 cycles	Negative underpredicts on average

The negative bias is consistent with cap-100 targets when some official endpoints have RUL above 100. The comparison confirms the need for an expanded validation horizon schedule before choosing a production target cap. It must not be used to retroactively select cap-125 or another candidate.

4 Competition-ready system components

- Sensor ingestion, integrity checks, causal rolling features, labelled RUL training, and saved reproducible artifacts.
- Compact CNN, LSTM, CNN-LSTM, TinyML-oriented, autoencoder, hybrid, and tabular model comparisons.
- Early-life autoencoder anomaly signal with a development-derived threshold. FD001 has no cycle-level anomaly labels, so no fabricated classification accuracy is claimed.
- Plain deterministic triage using RUL, anomaly persistence, data quality, and model version; it supports human review and prevents duplicate mock work orders.
- Streamlit dashboard driven by saved results, plus training-only data validation, descriptive drift indicators, and a two-version retraining simulation.

5 Recommendations

Use the cap-100 result as a controlled validation finding, not a final deployment claim. Add predeclared horizons above 100 cycles and repeat grouped validation before setting a production target cap. Then extend to FD002 and FD004 with operating-condition-aware preprocessing, calibrate intervals and triage limits with domain owners, and measure TinyML latency, RAM, quantization error, and update cost on the actual edge hardware.

Validation status

The audit generated data/pipeline/metric/protocol/baseline/ablation reports, split and endpoint files, per-candidate predictions, OOF error figures, a serialized validation-selected model, and MLOps artifacts. The test suite completed with 17 passing tests and one expected optional-dependency skip.